

Statement of Work
Astro P25 Land Mobile Radio System Support

Requesting Laboratory / Division / Group:
Communications Technology Laboratory
Public Safety Communications Research Division
Network Operations Group (671.01)

I. BACKGROUND:

The Public Safety Communications Research (PSCR) Division provides research, development, testing, and evaluation to foster nationwide communications interoperability. In order to test accurately, monitor and measure Land Mobile Radio (LMR) and Long-Term Evolution (LTE) signals, PSCR requires extended support of its Motorola P25 ASTRO LMR system. This system enables PSCR to conduct in-depth research, development, testing and evaluation of LMR and LTE systems and interoperability scenarios.

NIST PSCR acquired its Motorola Solutions Astro P25 Land Mobile Radio System in 2018-2019. The current system is on System Release A7.18 which reached end of support by Motorola in 2025. Motorola will no longer provide service support for the system.

PSCR experienced an extended outage in 2023 that required Motorola personnel to restore operation under the support contract which was in force at that time. A second complete outage occurred in October 2025 and was recovered by internal resources. These outages, with impacts to internal R&D, industry research partners, and the FirstNet Boulder lab, emphasized the critical importance of securing extended support from an organization with specific experience providing support and preventive maintenance services for Motorola systems on release A7.18, together with associated Motorola mobile and portable radios.

II. PURPOSE:

PSCR requires continuation of support for the Motorola LMR system and associated Motorola mobile and portable radios in order to continue research in the CTL Innovation Laboratory as well as technology demonstrations with stakeholders, first responders and federal entities, at the PSCR and FirstNet Boulder laboratories.

The support contract shall include:

- Preventive maintenance for infrastructure, fixed network equipment, and subscriber devices
- Assistance in identifying replacement components from secondary markets
- Remote technical support
- Dispatch service and on-site infrastructure response.

Place of Performance:

Contractor shall provide support services through remote and onsite work as provided under Section III. All onsite work shall be performed on the NIST campus in Boulder, Colorado: Building 3 Second Floor CTL Innovation Lab and Green Mountain RF site.

Regular Business Hours:

Regular business hours are Monday through Friday, 8:00 am to 5:00 pm Local Time, excluding Federal holidays and NIST closures.

Period of Performance/Delivery Date:

Base Period: Twelve (12) months After Release of Order (ARO)

Option Period 1: Twelve (12) months

Option Period 2: Twelve (12) months

III. SPECIFIC REQUIREMENTS:

Task 1 - Preventive Maintenance (PM)

The contractor shall perform one annual PM for the covered equipment (see Exhibit A) to ensure that infrastructure, fixed network equipment, and subscriber devices remain compliant with the original manufacturer's specifications through scheduled inspection and service:

- 1.1 Physically inspect all covered equipment.
- 1.2 Remove dust and foreign substances.
- 1.3 Clean air filters.
- 1.4 Measure, record, align, and adjust components to meet manufacturer specifications.
- 1.5 Update firmware to the latest applicable version.
- 1.6 Perform PM for covered subscriber units: update firmware, tune and alignment. PM includes one alias change and radio template update per year at time of PM.
- 1.7 Assist with identification and sourcing of older, broken or obsolete system components that may still be available on the secondary market.

Task 2 - Remote Technical Support

The contractor shall provide expert consultation via telephone for advanced network troubleshooting:

- 2.1 Provide access to experienced communications technologists.
- 2.2 Provide access to technicians for wiring or error questions.
- 2.3 Provide escalation tracking and case management.
- 2.4 Provide measurable metrics to ensure network performance and longevity.
- 2.5 Ensure compliance with contracted response times (see Section VI).

Task 3 - Dispatch Service and On-site Infrastructure Response

contractor shall:

- 3.1 Dispatch qualified technicians to respond on-site to diagnose and restore communication system and network issues.
- 3.2 Provide automated case and escalation tracking.
- 3.3 Ensure compliance with contracted response times (see Section VI).
- 3.4 Provide historical activity records for system performance transparency.

Task 4 - Safety-Related Equipment

The Contractor shall:

Provide all safety-related equipment required to perform the work under this award, including personal protective equipment (PPE) and other equipment that is required or appropriate to protect contractor personnel and other personnel on the NIST campus. Contractor-provided personal protective equipment shall include, at a minimum, hard hats, safety shoes, safety glasses (prescription or standard), respiratory protection (powered air purifying respirator or full-face mask respirator), high visibility clothing and gloves that meet applicable OSHA standards for PPE.

Task 5 - Training

The Contractor shall:

Provide safety-related training required by applicable OSHA, state and local regulations or standards to affected contractor personnel (including subcontractor personnel) that perform work under this award. Such training may include, for example, training for proper use of PPE, respirator use, fall protection, crane operation, forklift operation; electrical safety, or other safety-related topics.

Maintain records of safety training provided to contractor personnel (including subcontractor personnel) that perform work on this award. The contractor shall make those records available to the contracting officer, contracting officer's representative for the award, or other authorized personnel within 7 days of receipt of their written request, and annually thereafter or upon request.

Require contractor personnel (including subcontractors) to attend required NIST-specific health or safety training that is required by the award.

Task 6 - Health Hazard Assessments and Medical Evaluations

The Contractor shall: Provide required health hazard exposure assessments, medical evaluations and screenings, and exposure monitoring required by applicable OSHA requirements.

Ensure PII information is protected.

Task 7 - Homeland Security Presidential Directive (HSPD)-12 (Required Task)

The Contractor (and/or any subcontractor) and its employees who require physical access to NIST facilities or logical access to NIST IT networks or systems must comply with Homeland Security Presidential Directive (HSPD)-12, Policy for a Common Identification Standard for Federal Employees and Contractors; OMB M-05-24; OMB M-19-17; FIPS 201, Personal Identity Verification (PIV) of Federal Employees and Contractors; NIST HSPD-12 policy; and Executive Order 13467, Part 1 §1.2.

The Contractor (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, have the ability to access, or host and/or maintain a government information system(s).

The roster must be submitted to the COR and/or CO within 14 days of the effective date of this contract. Any revisions to the roster as a result of staffing changes must be submitted within 14 days of the change. The COR will notify the Contractor of the appropriate level of investigation required for each staff member. If the employee is filling a new position, the Contractor must provide a position description, and the Government will determine the appropriate suitability level.

Position Sensitivity Designations Requirement

All Contractor (and/or any subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR).

The following position sensitivity designation levels apply to this solicitation/contact:

IV. GENERAL REQUIREMENTS

Government Responsibilities:

1. NIST is responsible for taking reasonable steps to avoid unnecessary damage to equipment.
2. NIST is responsible for monitoring system operation, resolving issues and alarms internally when possible, and reporting issues and failures to the Contractor for resolution, as well as capturing failure mode data, diagnostic data and logs (upon request the Contractor).
3. NIST is responsible for maintaining system backups
4. NIST is responsible for transporting radios which are unavailable during scheduled PMs to a vendor's location for service.

Inspection and Acceptance

In addition to the inspection and acceptance terms articulated in the specific FAR clause that allows the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

Performance Tests:

Test	Description	Standard
1	Check Unified Event Manager (UEM) following all system work.	All alarms are cleared or have mitigation plans (TPOC may waive if due to equipment failure).
2	Verify Phase 1 and Phase 2 radio calls on all talk groups from NIST Boulder Building 3 CTL Innovation Lab.	All calls are completed successfully.
3	Verify interoperability calls using one or more ISSI connections.	All configured ISSI connections operate successfully (subject to foreign system operability).
4	After calibration or adjustments, perform system drive testing for basic calls or Bit Error Rate Testing (BERT).	Phase 1 results are comparable with system baselines (i.e., no significant degradation in coverage footprint or performance).

NIST may choose at its discretion to forego all or part of acceptance testing.

The Government will test, inspect, and accept or reject the equipment within ten (10) days of completion of system maintenance activities unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

REQUIREMENTS FOR CONTRACTOR PERSONNEL

(1) Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

(2) Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

(3) Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

(4) Vehicle Registration: All Contractor employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass shall be displayed on the vehicle in accordance with NIST Security Office instructions.

V. Schedule of Deliverables:

Deliverable	Description	Format	Due Date	Quantity
1	System Preventive Maintenance Report	Word, Excel, PDF	15 days after system PM	1
2	Mobile and Portable Radio Preventive Maintenance Report (firmware upgrade details, conformance results)	Word, Excel, PDF	15 days after radio PM	1

Standards of Acceptance:

The NIST Point of Contact (POC) shall review all the above deliverables and respond with an acceptance or request for revision email to the Contractor POC within 15 days of receipt of deliverable. This section applies to deliverables which must be reviewed and approved for acceptance (e.g., draft plans, drawings, etc.)

VI. SERVICE LEVEL AGREEMENT SUPPORT PLAN

Contractor will respond to service requests according to the following plan:

Severity	Description	Phone Response Time	On-Site Response Time
1	Catastrophic failure Impeding operations or jeopardizing safety	4 hours	Same day (after hours at T&M rates)
2	Operational degradation without safety risk	4 hours (calls after 2 p.m. responded next business day at 0800)	Same day or next business day
3	Minor performance issues, no significant impact	4 hours (calls after 2 p.m. responded to next business day at 0800)	Next business day

Exhibit A: Covered Equipment List

Motorola A7.18 M1 Core

M1 Core

VMS Servers, GGSN, GGM Gateways and LAN switches

One (1) license of each: Unified Event Manager (UEM), Unified Network Configurator (UNC), Provisioning Manager, Historical Reports, Zone Watch

One (1) Trunking Site license

One (1) Conventional Site license
P25 FDMA and TDMA Operation
500 Subscriber licenses
Trunking Integrated Voice and Data (IV&D)
Conventional Integrated Voice and Data (IV&D)
Dynamic Dual Mode (DDM)

One (1) ESS with 4 Trunking GTR 8000 repeaters – 700/800 MHz

Two (2) Conventional GTR 8000 repeaters – 700/800 MHz

Combining equipment, including antennas and transmission line

One (1) Network Management workstation

Customer Enterprise Network (CEN) to include the following:

Border Gateway, Firewall, and LAN switch

Packet Data Encrypted Gateway (PDEG)

GPS Server and Mapping Client

Intelligent Middleware Server (IMW)

Affiliation Display

ISSIs

VMS Server

Two (2) ISSI licenses

Motorola Mobile and Portable Radios

Quantity	Model	Description
2	APX1000	700/800 MHz
1	APX4000	VHF/UHF/700/800/900
1	APX6000	UHF
1	APX8000	700 MHz
1	APX8000	700/800
2	APX8000	VHF/UHF/700
1	APX8500	700/UHF
1	APX8500	700/800